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LADDER WITH LIFTING AND LOWERING HANDRAIL

Abstract

A ladder with a lifting and lowering handrail includes a ladder body and a handrail frame configured to be lifted and lowered on the ladder body. Two front leg tubes and two rear leg tubes are pivotally connected to form a foldable structure. The handrail frame includes a lateral rod and two side rods. Two pivoting sleeves respectively cover tops of the two front leg tubes. Tops of the two rear leg tubes are pivotally connected to the two pivoting sleeves. C-shaped openings of the two side rods are located on inner sides of the two side rods and cover outer sides of the two front leg tubes. The two side rods are configured to slide on the two pivoting sleeves. A lower end of at least one of the two side rods is disposed with a locking device configured to control the handrail frame to be lifted and lowered.

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Background/Summary

RELATED APPLICATIONS

[0001] This application claims priority to Chinese patent application number 202420466228.9, filed on Mar. 11, 2024. Chinese patent application number 202420466228.9 is incorporated herein by reference.

FIELD OF THE DISCLOSURE

[0002] The present disclosure relates to a folding ladder and in particular to a ladder with a lifting and lowering handrail.

BACKGROUND OF THE DISCLOSURE

[0003] Folding ladders have the advantages of being lightweight and easy to store, making them a common household tool sold in large supermarkets. Depending on whether the ladder has handrails, they are generally divided into two types: one type is the V-shaped ladder without handrails, and the other type is the inverted Y-shaped ladder with handrails. Handrails serve a protective role and help users stand more steadily on the highest step.

[0004] However, most existing handrails are directly fixed to leg tubes of the ladder, either as a one-piece structure, a welded structure, or a bolt-secured structure. That is, the handrail is a fixed structure, which, while considering convenience for storage, results in the handrail being relatively low in height. When a person is on the highest step of the ladder, the handrail may not be very practical due to its limited height. Furthermore, the fixed nature of the handrail limits the storage space, which can negatively affect normal use of the ladder.

BRIEF SUMMARY OF THE DISCLOSURE

[0005] The technical problem to be solved by the present disclosure is to provide a ladder with a lifting and lowering handrail.

[0006] In order to solve the above technical problems, the present disclosure provides the ladder with the lifting and lowering handrail. The ladder comprises a ladder body and a handrail frame sleeve on the ladder body and configured to be lifted and lowered. The ladder body comprises two front leg tubes, two rear leg tubes, and two pivoting sleeves, and the two front leg tubes and the two rear leg tubes are pivotally connected to form a foldable inverted Y-shaped structure. The handrail frame comprises a lateral rod and two side rods symmetrically arranged on a left and right direction, and upper ends of the two side rods are connected to two ends of the lateral rod. The two pivoting sleeves respectively cover tops of the two front leg tubes, and tops of the two rear leg tubes are respectively pivotally connected to the two pivoting sleeves. Cross sections of the two side rods are C-shaped cross sections, and C-shaped openings of the C-shaped cross sections are located on inner sides of the two side rods. The C-shaped openings of the two side rods cover outer sides of the two front leg tubes, and the two pivoting sleeves comprise position-providing grooves. The two side rods are configured to slide on the position-providing grooves, and a lower end of at least one of the two side rods is disposed with a locking device. The locking device is configured to be a lifting-and-lowering adjusting switch for controlling the handrail frame to be lifted and lowered.

[0007] In a preferred embodiment, an upper end surface of each of the two pivoting sleeves comprises a step surface, and outer sides of the two pivoting sleeves comprise the position-providing grooves. When the handrail frame is retracted, the lateral rod abuts the step surface.

[0008] In a preferred embodiment, each of the two pivoting sleeves comprises a first sleeve and a second sleeve, and the first sleeve is sleeved on a corresponding one of the two front leg tubes. The second sleeve is sleeved on a corresponding one of the two rear leg tubes, and the second sleeve comprises a space for the corresponding one of the two rear leg tubes to rotate.

[0009] In a preferred embodiment, the corresponding one of the two rear leg tubes is rotatably

connected in the second sleeve through a rivet.
[0010] In a preferred embodiment, the locking device comprises a hand-held sleeve sleeved on the lower end of the at least one of the two side rods, a button pivotally connected to the hand-held sleeve, at least one locking hole arranged on a side of a corresponding one of the two front leg tubes, and a locking head disposed on the button and configured to correspond to the at least one locking hole. The locking head is configured to be disposed in or separated from the at least one locking hole driven by the button.
[0011] In a preferred embodiment, the hand-held sleeve is buckled to the at least one of the two side rods, and a middle part of the button is pivotally connected to the hand-held sleeve. The button is located on a rear side of the hand-held sleeve, and the locking head is disposed on an inner side of a lower end of the button. An elastic member is disposed on an inner side of an upper end of the button, and the locking head is configured to be disposed in the at least one locking hole driven by the elastic member.
[0012] In a preferred embodiment, a part of the button that forms the locking head extends out of the hand-held sleeve and forms an interlocked connection with the at least one locking hole on the corresponding one of the two front leg tubes.
[0013] In a preferred embodiment, the at least one locking hole is at least one rectangular hole, and the locking head is a protruding surface. The locking head comprises a protruding part and two buckling edges, and the two buckling edges are buckled in the button. The protruding part is disposed outside the button and is configured to be disposed in the at least one rectangular hole.
[0014] In a preferred embodiment, the at least one locking hole is a plurality of locking holes, and the plurality of locking holes are spaced apart along a length direction of the corresponding one of the two front leg tubes.
[0015] In a preferred embodiment, the handrail frame is U-shaped, and the lower end of each of the two side rods is disposed with the locking device.
[0016] Compared with the existing techniques, the technical solution has the following advantages.
[0017] 1. The two front leg tubes and the two rear leg tubes are connected through two pivoting sleeves. Cross-sections at connection ends of the two front leg tubes and the two front leg tubes lie in a same plane, making an overall ladder look simple, reliable, and aesthetically pleasing. [0018] 2. The ladder body is disposed with the lifting and lowering handrail, allowing the ladder to have the handrail frame while still being foldable for storage. The lifting and lowering handrail can also meet the needs of different heights for handrail use, enhancing functionality. [0019] 3. The at least one rectangular hole, combined with a locking head with a protruding surface, is more stable than a conventional round locking hole. The at least one rectangular hole can better lock the locking head, forming a more stable locking structure, avoiding a problem of the locking device coming off, and improving the ladder's safety during use.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0020] FIG. **1** is a perspective view of a ladder in a use state in a preferred embodiment of the present disclosure.
[0021] FIG. **2** is a perspective view of the ladder, including a cross-sectional view on a pivoting sleeve of the ladder in the preferred embodiment of the present disclosure.
[0022] FIG. **3** is an enlarged view of section A in FIG. **1**.
[0023] FIG. **4** is an enlarged view of section B in FIG. **2**.
[0024] FIG. **5** is a perspective view of part of the ladder, including a cross-sectional view of a locking device in the preferred embodiment of the present disclosure.
[0025] FIG. **6** is a perspective view of a locking hole and a locking head being inserted into each

other in the preferred embodiment of the present disclosure.

[0026] FIG. **7** is a perspective view of the locking hole in the preferred embodiment of the present disclosure.

[0027] Explanation of the reference numerals: **1**. ladder; **11**. front leg tube; **111**. locking hole; **12**. rear leg tube; **13**. pivoting sleeve; **131**. Position-providing groove; **132**. first sleeve; **133**. second sleeve; **134**. rivet; **135**. step surface; **2**. handrail frame; **21**. lateral rod; **22**. side rod; **3**. locking device; **31**. hand-held sleeve; **32**. button; **33**. locking head; and **34**. elastic member.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0028] The technical solutions of the embodiments of the present disclosure will be described in detail below with reference to the drawings. It should be understood that the embodiments described are only part of the embodiments of the present disclosure, and not all possible embodiments. All other embodiments that would be apparent to a person skilled in the art based on the teachings of the present disclosure, without requiring any inventive step, are within the scope of the present disclosure's protection.

[0029] In the description of the present disclosure, it should be noted that terms such as “upper”, “lower”, “inner”, “outer”, “top end”, “bottom end”, and the like, refer to positions or positional relationships based on the orientations shown in the drawings. These terms are used for convenience in describing the present disclosure and simplifying the description, and should not be interpreted as limiting the present disclosure. The device or element in question does not necessarily have to be constructed or operated in a specific orientation or position. Additionally, the terms “first”, “second”, and similar terms are used for descriptive purposes only and do not imply any relative importance or ranking.

[0030] In the description of the present disclosure, unless otherwise clearly stated or limited, terms such as “installed”, “provided with”, “sleeved on”, “connected”, etc., should be understood in their broadest sense. For example, “connection” can refer to a wall-mounted connection, a detachable connection, or an integral connection. It may refer to a mechanical or electrical connection, direct or indirect connection, or an internal connection between components. The specific meaning of these terms will be understood by those skilled in the art based on the particular context of the disclosure.

[0031] Referring to FIG. **1** to FIG. **7**, the present embodiment provides a ladder **1** with a lifting and lowering handrail. The ladder **1** comprises two front leg tubes **11**, two rear leg tubes **12**, and two pivoting sleeves **13**. The two front leg tubes **11** and the two rear leg tubes **12** are pivotally connected to form a foldable inverted Y-shaped structure. The two pivoting sleeves **13** respectively cover tops of the two front leg tubes **11**, and tops of the two rear leg tubes **12** are respectively pivotally connected to the two pivoting sleeves **13**. Specifically, an upper end surface of each of the two pivoting sleeves **13** is a flat surface. Each of the two pivoting sleeves **13** comprises a first sleeve **132** and a second sleeve **133**. The first sleeve **132** is sleeved on a corresponding one of the two front leg tubes **11**, and the second sleeve **133** is sleeved on a corresponding one of the two rear leg tubes **12**. The corresponding one of the two rear leg tubes **12** is pivotally connected in the second sleeve **133** through a rivet **134**, and the second sleeve **133** comprises a space for the corresponding one of the two rear leg tubes **12** to rotate. The two front leg tubes **11** and the two rear leg tubes **12** are folded and unfolded by using the two pivoting sleeves **13**, so that a connection between the two front leg tubes **11** and the two rear leg tubes **12** is more stable and unfolding and folding of the ladder **1** is more convenient. At the same time, the upper end surface, which is flat, of each of the two pivoting sleeves **13** is convenient for supporting the ladder **1**, and is beautiful and neat.

[0032] A handrail frame **2** is U-shaped, and the handrail frame **2** comprises a lateral rod **21** and two side rods **22** symmetrically arranged in a left and right direction. Upper ends of the two side rods **22** are connected to two ends of the lateral rod **21**. Cross sections of the two side rods **22** are C-shaped cross sections, and C-shaped openings of the C-shaped cross sections are located on inner sides of the two side rods **22**. The inner sides refer to opposite surfaces of the two side rods **22** facing an inside of the ladder **1**. The C-shaped openings of the two side rods **22** cover outer sides of the two front leg tubes **11**. Outer sides of the two pivoting sleeves **13** comprise position-providing grooves **131**, and the two side rods **22** are configured to slide on the position-providing grooves **131**. The C-shaped openings of the two side rods **22** can provide a space for the two side rods **22** and the two pivoting sleeves **13** and avoid interference between the handrail frame **2** and the two pivoting sleeves **13** during a lifting and lowering process. The C-shaped openings of the two side rods **22** cover the outer sides of the two front leg tubes **11**, so that the handrail frame **2** can be more stable during sliding and convenient for disassembly and assembly. The ladder **1** looks simple and reliable as a whole.

[0033] To make the ladder **1** and the handrail frame **2** more beautiful as a whole after being folded, the upper end surface of each of the two pivoting sleeves **13** comprises a step surface **135**, and an upper end protrusion of the second sleeve **133** on which the rear leg tube **12** is installed is higher than the step surface **135**. When the handrail frame **2** is retracted, the lateral rod **21** abuts the step surface **135**, and an upper end surface of the lateral rod **21** is exactly flush with the upper end protrusion of the second sleeve **133**, forming a flat structure, so that the ladder **1** is more neat and simple as a whole after being folded.

[0034] To effectively perform locking of the handrail frame **2** after being lifted and lowered, a lower end of each of the two side rods **22** is disposed with a locking device **3**, and the locking device **3** is configured to be a telescopic adjusting switch for controlling the handrail frame **2** to be lifted and lowered. The locking devices **3** are disposed on two sides of the ladder **1** to make the locking of the handrail frame **2** after being lifted and lowered more stable and firm.

[0035] The locking device **3** comprises a hand-held sleeve **31** sleeved on the lower end of a corresponding one of the two side rods **22**, a button **32** pivotally connected to the hand-held sleeve **31**, at least one locking hole **111** arranged on a side of the corresponding one of the two front leg tubes **11**, and a locking head **33** disposed on the button **32** and configured to correspond to the at least one locking hole **111**. The locking head **33** can be inserted into or removed from the at least one locking hole **111** driven by the button **32**.

[0036] A middle part of the button **32** is pivotally connected to the hand-held sleeve **31**, and the locking head **33** is disposed on an inner side of a lower end of the button **32**. An elastic member **34** is disposed on an inner side of an upper end of the button **32**. The locking head **33** can be inserted into the at least one locking hole **111** driven by the elastic member **34**.

[0037] When in use, the locking head **33** is inserted into the at least one locking hole **111** of the corresponding one of the two front leg tubes **11** driven by the elastic member **34**, so that the two side rods **22** of the handrail frame **2** and the two front leg tubes **11** of the ladder **1** are interlocked with each other, so as to achieve a purpose of fixing the handrail frame **2**. When a height of the handrail frame **2** needs to be adjusted, the button **32** is pressed, and the upper end of the button **32** presses the elastic member **34** to enable the lower end of the button **32** to rotate and tilt up, so that the locking head **33** can be separated from the at least one locking hole **111**, thereby releasing the interlocking connection of the handrail frame **2**, and the handrail frame **2** can be lifted and lowered.

[0038] To allow the handrail frame **2** to have different height adjustments, the least one locking hole **111** is a plurality of locking holes **111** arranged on the side of the corresponding one of the two front leg tubes **11**. The plurality of locking holes **111** are spaced apart along a length direction of the corresponding one of the two front leg tubes **11**. The plurality of locking holes **111** enable the handrail frame **2** to have a function of different height adjustment. The handrail frame **2** can be quickly lowered when stored, and the height of the handrail frame **2** can be adjusted according to requirements of a user when in use, thereby improving safety of the ladder **1**. The hand-held sleeve **31** is buckled to the corresponding one of the two side rods **22**, and the button **32** is disposed at a rear side of the corresponding one of the two side rods **22**. In this way, while the hand-held sleeve **31** is held and the button **32** is pressed, the handrail frame **2** can be better held for lifting and lowering adjustment.

[0039] To simplify a locking structure of the locking head **33** and the at least one locking hole **111**, a part of the button **32** that forms the locking head **33** extends out of the hand-held sleeve **31** and forms an interlocked connection with the at least one locking hole **111** on the corresponding one of the two front leg tubes **11**. The part of the button **32** that forms the locking head **33** is directly interlocked with the at least one locking hole **111** on the corresponding one of the two front leg tubes **11**, and an operation of pressing the button **32** can be more flexible and more convenient. In addition, when designing the structure, it is only necessary to provide the at least one locking hole **111** of the corresponding one of the two front leg tubes **11**, and there is no need to provide an avoidance hole on the hand-held sleeve **31** and the corresponding one of the two side rods **22**. The structure is more concise, and the locking head **33** is simpler to be inserted into the at least one locking hole **111**. There is no need for multi-hole alignment (i.e., in which the avoidance hole needs to be aligned with the at least one locking hole **111**), which can avoid a phenomenon in which the locking head **33** falls out and cannot be locked due to the multi-hole alignment.

[0040] To facilitate the operation of pressing the button **32**, the elastic member **34** is configured as a spring. A position of the button **32** corresponding to the elastic member **34** is disposed with a positioning pin **321**, and a position of the hand-held sleeve **31** corresponding to the elastic member **34** comprises a mounting hole **311**. A first end of the elastic member **34** is fixed in the mounting hole **311**, and a second end of the elastic member **34** is inserted into the positioning pin **321**. Movement of the first end and the second end of the elastic member **34** is limited, and a structure is more stable.

[0041] Alternatively, to make locking of the locking device **3** safer, the at least one locking hole **111** is a rectangular hole, and the locking head **33** is a protruding surface. The locking head **33** comprises a protruding part **331** and two buckling edges **332**, and the two buckling edges **332** are buckled in the button **32**. The protruding part **331** is disposed outside the button **32** and is configured to be inserted into the rectangular hole. Through the rectangular hole, the locking head **33** with a protruding surface can be inserted into the rectangular hole. Compared with a structure of a conventional circular locking hole, the rectangular hole can better lock the locking head **33**, forming a more stable locking structure, which can avoid a phenomenon of the locking device **3** being disengaged and improve the safety performance of the ladder **1**.

[0042] The above description represents only a preferred specific embodiment of the present disclosure and is not intended to limit its design concept. Any minor modifications made to the present disclosure based on this concept by those skilled in the art within the technical scope disclosed herein shall be deemed acts of infringing upon the protection scope of the present disclosure.

Claims

1. A ladder with a lifting and lowering handrail, comprising: a ladder body, and a handrail frame sleeve on the ladder body and configured to be lifted and lowered, wherein: the ladder body comprises two front leg tubes, two rear leg tubes, and two pivoting sleeves, the two front leg tubes and the two rear leg tubes are pivotally connected to form a foldable inverted Y-shaped structure, the handrail frame comprises a lateral rod and two side rods symmetrically arranged on a left and right direction, upper ends of the two side rods are connected to two ends of the lateral rod, the two pivoting sleeves respectively cover tops of the two front leg tubes, tops of the two rear leg tubes are respectively pivotally connected to the two pivoting sleeves, cross sections of the two side rods are C-shaped cross sections, C-shaped openings of the C-shaped cross sections are located on inner sides of the two side rods, the C-shaped openings of the two side rods cover outer sides of the two front leg tubes, the two pivoting sleeves comprise position-providing grooves, the two side rods are configured to slide on the position-providing grooves, a lower end of at least one of the two side rods is disposed with a locking device, and the locking device is configured to be a lifting-and-lowering adjusting switch for controlling the handrail frame to be lifted and lowered.

2. The ladder with the lifting and lowering handrail according to claim 1, wherein: an upper end surface of each of the two pivoting sleeves comprises a step surface, outer sides of the two pivoting sleeves comprise the position-providing grooves, and when the handrail frame is retracted, the lateral rod abuts the step surface.

3. The ladder with the lifting and lowering handrail according to claim 1, wherein: each of the two pivoting sleeves comprises a first sleeve and a second sleeve, the first sleeve is sleeved on a corresponding one of the two front leg tubes, the second sleeve is sleeved on a corresponding one of the two rear leg tubes, and the second sleeve comprises a space for the corresponding one of the two rear leg tubes to rotate.

4. The ladder with the lifting and lowering handrail according to claim 3, wherein: the corresponding one of the two rear leg tubes is rotatably connected in the second sleeve through a rivet.

5. The ladder with the lifting and lowering handrail according to claim 1, wherein: the locking device comprises a hand-held sleeve sleeved on the lower end of the at least one of the two side rods, a button pivotally connected to the hand-held sleeve, at least one locking hole arranged on a side of a corresponding one of the two front leg tubes, and a locking head disposed on the button and configured to correspond to the at least one locking hole, and the locking head is configured to be disposed in or separated from the at least one locking hole driven by the button.

6. The ladder with the lifting and lowering handrail according to claim 5, wherein: the hand-held sleeve is buckled to the at least one of the two side rods, a middle part of the button is pivotally connected to the hand-held sleeve, the button is located on a rear side of the hand-held sleeve, the locking head is disposed on an inner side of a lower end of the button, an elastic member is disposed on an inner side of an upper end of the button, and the locking head is configured to be disposed in the at least one locking hole driven by the elastic member.

7. The ladder with the lifting and lowering handrail according to claim 5, wherein: a part of the button that forms the locking head extends out of the hand-held sleeve and forms an interlocked connection with the at least one locking hole on the corresponding one of the two front leg tubes.

8. The ladder with the lifting and lowering handrail according to claim 5, wherein: the at least one locking hole is at least one rectangular hole, the locking head is a protruding surface, the locking head comprises a protruding part and two buckling edges, the two buckling edges are buckled in the button, and the protruding part is disposed outside the button and is configured to be disposed in the at least one rectangular hole.

9. The ladder with the lifting and lowering handrail according to claim 5, wherein: the at least one locking hole is a plurality of locking holes, and the plurality of locking holes are spaced apart along a length direction of the corresponding one of the two front leg tubes.

10. The ladder with the lifting and lowering handrail according to claim 1, wherein: the handrail frame is U-shaped, and the lower end of each of the two side rods is disposed with the locking device.